

should be used since every semiconductor has its own specifications. Pin configurations should be scrupulously followed. Each pin in a semiconductor is defined. Even equivalents of semiconductors may have different pin configurations.

Beginners are strongly recommended to use solderless breadboards so that necessary changes in the connections can be made easily. Also, the same components can be reused for other circuits.

Make sure that components are inserted upright and firmly in the breadboard. Components with moving parts like relays, gang capacitors and potentiometers should be mounted firmly on flat and firm surfaces so that their connecting wires do not get uprooted from the breadboard during experimentation.

Troubleshooting

Soldering is an art that needs a lot of skill and practice to master. An electronic circuit is seldom one unit unless it is a very basic circuit. Usually it consists of a number of stages joined in a logical sequence. Each stage should be checked independently.

Many schematics show voltage levels with respect to ground at vulnerable points in a circuit. This is helpful for checking the working of each individual stage.

Before powering the circuit, the battery or power supply connections must be checked. A reverse polarity may instantly damage the semiconductors. For safety, a rectifier diode may be used in series with the positive supply. Electrolytic capacitors, if wrongly connected, may get damaged instantly or, in rare cases, may even explode on powering.

A component, particularly a semiconductor, in a rare case, may get hot to the extent that it may burn itself. This is accidental and may occur due to wrong connections, reverse polarity, high voltage or a short connection in a circuit.

One more annoying culprit is a loose or an open connection. Use

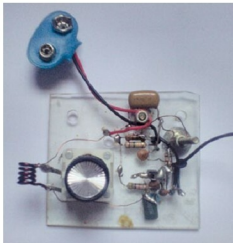


Fig. 3: FM transmitter assembled on a transparent acrylic sheet

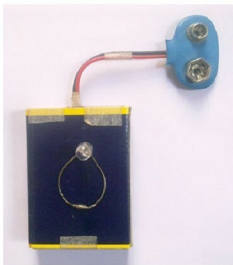


Fig. 4: Wireless LED using RF, boxed up in a matchbox

hook-up wires that fit firmly in the breadboard.

Take it for granted that the assembled circuit may not work in the maiden attempt. In such a case, power supply should be disconnected immediately and a thorough re-inspection of the connections should be carried out. Thus with patience, practice and perseverance, early experimenting after a long and unflagging pursuit can bloom into a lifetime hobby one can be proud of.

Advancement

In the past few decades, technology has advanced by leaps and bounds. Semiconductors have replaced vacuum tubes, silicon transistors have replaced germanium transistors, ICs have replaced discrete components, miniaturised components have re-



Fig. 5: Electrical-safety-ensuring device (pending patent)



Fig. 6: Astable multivibrator built around timer 555 with two LEDs flashing alternately



Fig. 7: LED directly on mains, boxed up in small transparent cases

placed bulky components, switched-mode power supplies (SMPs) have replaced transformer based power supplies and so on; not to mention the world of digital electronics and microprocessors.

Advancement of technology has changed the very outlook of electronics. A vista of ultra-modern electronics is anticipated in the near future. Nonetheless, the basics of electronics are going to entice the forthcoming generations of hobbyists and amateurs. It is no wonder, if amidst the extravaganza of ultra-modern electronics, a sprouting hobbyist will be seen toying with the most basic circuit like a relaxation oscillator. Conclusively, electronics as a hobby is likely to remain as live as a working circuit and that too for decades to come! ●